

Filon enclosure of the two integral classes

at $L = 0.3$: finite S is interval-PD, leftover C blocks the SATZ

PRIME.RDAGGER.FILON_ENCLOSURE.01 — round 483

Stefan Hamann

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Abstract

The r481 remainder was the pair of integrals $\int_0^T \sigma_A \hat{\psi}_i \hat{\psi}_j$ and $\int_T^\infty \sigma_A \hat{\psi}_i \hat{\psi}_j$ in the refined even Schur $S = B - CC^\top / c_{\text{tail}}$. This round encloses both. The Nyström sample Fourier transform is an explicit exponential sum; panel-linear Filon with a split M_2 remainder seals every $[0, T]$ entry (midpoints hit the r480/r481 trapezoid to 10^{-7}). The $[T, \infty)$ tail of the exponential sum is almost-periodic and is *not* used (Ci-sum to infinity yields $\lambda_{\min}(S) \sim -1.6$, killed). The decaying interpolant Fourier transform is the four-term endpoint integration-by-parts of the sinc extension; diagonal tails are positive and booked to B (A_+ side, sign bookkeeping explicit); off-diagonal tails are kept as signed values. Higham on the finite even 3×3 Schur with entry radius 10^{-4} gives $\mu \geq 1.19 \cdot 10^{-3}$ (odd $\mu \geq 0.103$). A would-be $c_* = \mu / (1 + \|C\|^2 / c_{\text{tail}}^2) \approx 1.68 \cdot 10^{-4}$ exists *if* C were the full operator coupling. It is not. Leftover $\|P_{\text{block}} A P_{\geq 12}\|_F = 2.76 \cdot 10^{-2}$ on $[0, T]$ (modes 12, 14) against an IBP-inflated budget $\|C_{\text{rest}}\| < 2.4 \cdot 10^{-4}$. Deficit factor ~ 115 . Verdict **REDUCED(finite-S-interval-PD; leftover-C-HS=2.74e-2-vs-budget-2.4e-4)**. Fourier-HS is not reused. Yoshida-Bombieri zone. No RH claim.

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Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. The finite Schur is interval-PD; leftover C blocks the SATZ. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$ (scramble-sensitive, vacuum). Plancherel $\hat{h}(t) = \int h(x) e^{itx} dx$, $\|h\|_2^2 = (2\pi)^{-1} \int |\hat{h}|^2$, $A = (2\pi)^{-1} \int \sigma_A |\hat{h}|^2$ with $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$. The r469/r479/r480 contract is in force: $t_* = 8$, $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$, $c_{\text{tail}} \geq 0.2408$, even block $\{\psi_0, \psi_2, \psi_4\}$ versus even tail $\{\psi_6, \psi_8, \psi_{10}\}$ (odd partner $\{\psi_1, \psi_3, \psi_5\}$ versus $\{\psi_7, \psi_9, \psi_{11}\}$).

2 Class 1: Filon on $[0, T]$

The Nyström interpolant sampled at Gauss-Legendre $n = 48$ has Fourier transform

$$\hat{\psi}_i(t) = \sum_{q=1}^{48} w_q \psi_i(x_q) e^{itx_q},$$

an exponential sum. No x -quadrature remainder attaches to this discrete object. Write $\alpha_{iq} = w_q \psi_i(x_q)$. Evenness gives

$$A_{ij}[0, T] = \frac{1}{\pi} \sum_{q,r} \alpha_{iq} \alpha_{jr} J_T(x_q - x_r), \quad J_T(\omega) = \int_0^T \sigma_A(t) \cos(\omega t) dt.$$

Linear interpolation of σ_A on panels, with $|\sigma_A''| \leq 36.8$ on $(0, 8]$ and $|\sigma_A''| \leq 0.018$ on $[8, 150]$ (finite-difference majorants, 15% pad), has per-band L^1 remainder $M_2 h^3 / 8 \cdot N_{\text{pan}}$. At $N_{\text{pan}} = (8000, 8000)$ one has $\text{rem}_{L^1} \leq 1.38 \cdot 10^{-4}$, hence $|\Delta A| \leq (\sum |\alpha_i|)(\sum |\alpha_j|) \text{rem}_{L^1} / \pi \leq 2.5 \cdot 10^{-5}$. Midpoints differ from the r480 trapezoid by at most $1.72 \cdot 10^{-7}$ on the even diagonal.

Lemma 1 (Filon on the compact interval). *Every even/odd $A_{ij}[0, 150]$ of the Nyström sample transform is enclosed to radius $2.5 \cdot 10^{-5}$. The no-tail even Schur has midpoint $\lambda_{\min}(S) = 5.692 \cdot 10^{-3}$, matching r480 ($5.686 \cdot 10^{-3}$).*

3 Class 2: decaying IBP tail

Exponential-sum Ci (killed). The product $\hat{\psi}_i \hat{\psi}_j$ of two exponential sums is a double trigonometric polynomial with frequencies $|x_q - x_{q'}| \leq 2L$. The $\omega = 0$ (diagonal $q = q'$) coefficient is $\text{DC}_{ij} = \pi^{-1} \sum_q \alpha_{iq} \alpha_{jq}$. On the diagonal $\text{DC}_{ii} > 0$ and $\int_T^\infty \sigma_A = +\infty$; dropping it would lower A_{ii} (safe for B). Off-diagonal DC_{ij} can have either sign ($-1.40 \cdot 10^{-3}$ for $(2, 4)$) and is not safe to drop. The oscillatory Ci-sum of the exponential sum, DC dropped, produces $\lambda_{\min}(S) \approx -1.62$. The exponential sum does not decay; it is not the $[T, \infty)$ Fourier transform of the interpolant.

Four-term IBP of the interpolant (kept). The sinc extension $\tilde{\psi}(x) = \lambda^{-1} \sum_j w_j K(x, x_j) \psi(x_j)$ is entire. For even ψ ,

$$\hat{\psi}(t) = \sum_{k=0}^3 2 \psi^{(k)}(L) \frac{\phi_k(tL)}{t^{k+1}} + R_4(t),$$

with phases $(\sin, \cos, -\sin, -\cos)$ and $|R_4(t)| \leq 2L \|\psi^{(4)}\|_\infty / t^4$. At $T_{\text{rem}} = 800$ the product remainder on C_{46} is $O(10^{-5})$. Binet: $\sigma_A(t) = \log(t/(2\pi)) - 1/(24t^2) + O(t^{-4})$, sealed by $|t^2 \rho| = 1/24$ on $[20, 800]$.

Sign bookkeeping. (i) Diagonal IBP tails are positive ($\text{diag } A_{\text{tail}} \approx (0.0078, 0.130, 0.266, 0.385, 0.500)$ even) and are booked to B (A_+ enlargement). (ii) Off-diagonal IBP tails are included as signed values, not dropped and not moved into the error. (iii) Exponential-sum $\omega = 0$ pairs are not integrated to infinity. (iv) Leftover C after ψ_{10} is not moved into B or into c_{tail} .

Lemma 2 (Finite IBP Schur). *Adding the four-term IBP tail on modes 0..11 keeps*

$$\lambda_{\min}(S_{\text{even}}) = 5.949 \cdot 10^{-3}, \quad \|C\|_{\text{even}} = 0.595,$$

$$\lambda_{\min}(S_{\text{odd}}) = 0.211, \quad \|C\|_{\text{odd}} = 0.631.$$

Higham with entry radius 10^{-4} yields $\mu_{\text{even}} \geq 1.196 \cdot 10^{-3}$ and $\mu_{\text{odd}} \geq 0.103$.

4 Error budget

Source	Even entry / op.	vs. margin
Filon M_2 rem $[0, 150]$	$2.5 \cdot 10^{-5}$	$\ll \mu$
IBP series rem at $T = 800$	$O(10^{-5})$ on C_{46}	$\ll \mu$
Trapezoid of IBP integrand	$< 5 \cdot 10^{-5}$	$\ll \mu$
Higham μ (radius 10^{-4})	$1.20 \cdot 10^{-3}$	—
Would-be c_* (leftover = 0)	$1.68 \cdot 10^{-4}$	—
Leftover $\ C_{\text{rest}}\ _F$ (modes 12, 14)	$2.76 \cdot 10^{-2}$	
IBP-inflated leftover budget	$2.4 \cdot 10^{-4}$	
Perturbation $(2\ C\ \ C_r\ + \ C_r\ ^2)/c_{\text{tail}}$	0.139	$116 \times \mu$

The exact deficit is the leftover coupling of the even block to ψ_{12}, ψ_{14} ($\lambda_{12}, \lambda_{14}$ sit at the Nyström floor $\sim 10^{-16}$). As an operator perturbation of S it is 0.139 against Higham $\mu = 1.20 \cdot 10^{-3}$. Modes 16+ have garbage jets and are not used as a pad.

5 Kills

Object	Round	Status
Fourier-HS majorant	r478	not reused
Crude $\kappa_{\text{high}} \leq 1.79 \cdot 10^{-2}$	r480	killed (≈ 0.29)
Integer Fourier ONB	r481	$Q_{0n} \sim 1/n$
σ^2 as $\ Au\ _{[-L, L]}$	r481	$\lambda_{\min}(S) \approx -10$
IBP envelope as pad	r481	0.76 on mode 4
Exp-sum Ci to ∞	r483	$\lambda_{\min}(S) \approx -1.62$

6 Scaling at $L = 0.8$

At $L = 0.8$ the prime range $n \leq e^{2L}$ is nonempty: P is a cosine sum with literal $\log n$ (scramble-sensitive, live). The exponential-sum structure of $\hat{\psi}$ survives; $\sigma_A - \sigma_P$ is still an explicit combination of a Digamma symbol and a finite cosine sum, so Filon and four-term IBP apply verbatim. What becomes more expensive: more frequencies $|x_q - x_{q'}| \leq 2L$, a larger M_2 of σ_P on $[0, t_*]$, and a rank-2 (not rank-3) P after the r469 pairing. The leftover- C obstruction is not cured by L ; it is a completeness statement about C , not about T .

7 What is not claimed

The finite interval-PD Schur is a lemma, not $\lambda_*(0.3) \geq c$. Assembling the SATZ would require $\|C_{\text{rest}}\| < 2.4 \cdot 10^{-4}$ (or a Loewner substitute that does not inflate $\|C\|$ to 0.60). Neither is available. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.